

Real-Time Fractional Tracking (R-TFT): φ -Weighted Mass Resonance Identifier

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Abstract

This paper introduces a φ -resonant framework for identifying mass resonance coherence across particle clusters, isotopic lattices, and complex fields using Real-Time Fractional Tracking (R-TFT). Rather than using classical frequency centroids or Fourier bins, we define resonance windows using φ -scaled bandwidths ($\Delta f_k = \varphi \cdot \Gamma_k / 2\pi$) and enforce recursive selection based on slice alignment metrics. Signals are retained only when coherence exceeds the golden threshold ($C_\varphi < 0.618$), allowing for dimensional convergence without static assumptions. Mass-cluster weighting is modulated by resonance stability and abundance symmetry, revealing hidden attractor states across quantum, stellar, and bioscalar domains. This tool serves as a universal resonance discriminator, identifying stable coherence bundles with minimal noise intrusion. Fully licensed under REL-2.0, all applications must remain peaceful, non-invasive, and consent-based.

1 Introduction

Detecting coherent mass resonance patterns remains a fundamental challenge across domains ranging from nuclear matter to stellar clusters. Traditional approaches lack real-time resolution and ethical safeguards. This work proposes a φ -aligned coherence filter based on R-TFT, designed to isolate stable resonance bundles via adaptive slice metrics and golden-ratio gating.

2 Method: Resonant Filtering with φ Bandwidths

Each resonance mode k is assigned a φ -weighted bandwidth:

$$\Delta f_k = \frac{\varphi \cdot \Gamma_k}{2\pi} \quad (1)$$

The system filters incoming signals through these dynamic windows and evaluates coherence via:

$$C_\varphi = |\text{Dim}(A) - \text{round}(\text{Dim}(A))| \quad (2)$$

Only if $C_\varphi < 0.618$ and $\dot{R}_{\text{clean}} > 0$ does the mass bundle qualify as resonant.

3 Mass Cluster Coherence and Weighting

Mass identifiers are grouped based on proximity in R-TFT slice space. Relative weighting is derived from:

- Clustered coherence ($\sum R_i$ stability)
- Abundance symmetry (population entropy)
- Dimensional embedding coherence (via φ -recursive fit)

This yields a prioritized output of mass-resonant identifiers tuned to dimensional feedback stability.

4 Applications

- Detection of stable isotopic shells
- Analysis of nuclear decay attractors
- Mass symmetry resonance in quark-gluon plasma
- Planetary mass band harmonics and star cluster coherence

5 Ethical Framework and REL-2.0

This system operates under REL-2.0 constraints. It may not be used to scan, track, or analyze living systems without full consent. All applications must remain transparent, peaceful, and scientifically motivated. Violations of recursive ethics constitute a resonance breach.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>.

6 Conclusion

Mass resonance is not merely a matter of frequency, it is a recursive coherence phenomenon. Using golden-ratio bandwidth logic and real-time feedback gating, this paper offers a universal identification framework across physical systems. All resonance must remain within ethical containment.